Analysis of PDEs

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Lecture 1

1 Basics

Let $\mathcal{U} \subseteq \mathbb{R}^n$ be an open set. A *PDE of order k* is an expression of the following form

$$F(x, u, Du, \dots, D^k u) = 0 \tag{1}$$

where $u : \mathcal{U} \to \mathbb{R}$ is the unknown, $F : \mathcal{U} \times \mathbb{R} \times \mathbb{R}^n \times \cdots \times \mathbb{R}^{n^k}$ is a general function. We say that u is a *classical solution of* eq. (1) if u satisfies eq. (1) in $\mathcal{U}$.

We may also consider the case where $u(x) \in \mathbb{R}^p$ and $F \in \mathbb{R}^q$, and we call this a *system of PDEs*.

1.1 Examples of PDEs

Example 1.1 (ODE system)

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One example of a model used in mathematical biology is the ODE system

$$\frac{\mathrm{d}u}{\mathrm{d}t} = f(u, v)$$
$$\frac{\mathrm{d}v}{\mathrm{d}t} = g(u, v)$$

Example 1.2 (Laplace's equation)

$$\Delta u = \sum_{i=1}^{n} \partial_i^2 u = 0$$

The Laplacian is an *averaging* operator.

Example 1.3 (Heat equation)

$$u_t = D\Delta u$$

This is also called the diffusion equation, D is called the diffusion constant.

Example 1.4 (Navier-Stokes)

The Navier-Stokes equations in fluid dynamics is

$$u_t = v\Delta u - u \cdot \nabla u - \nabla \rho + f$$
$$\operatorname{div}(u) = 0$$

Example 1.5 (Transport equation)

The transport equation is

$$u_t + v u_x = 0$$

where v is a constant, corresponding to the velocity. A modification is the *advection-diffusion* equation,

$$u_t + v \cdot \nabla u = D\Delta u + f$$

Example 1.6 (Poisson equation)

$$\Delta u = f$$

Describes electric field due to some charge, or Newtonian gravity.

Example 1.7 (Wave equation)

$$\Box u = -u_t t + c^2 \Delta u = 0$$

This models sound waves, seismic waves, ...

Example 1.8 (KdV equation)
This equation admits *soliton* solution.

$$u_t + \partial_x^3 u - 6u\partial_x u = 0$$

Example 1.9 (Maxwell equations)

$$\begin{aligned} \operatorname{div}(E) &= \rho \\ \operatorname{div}(B) &= 0 \\ \partial_t E &= \nabla \times B = J \\ \partial_t B &= -\nabla \times F \end{aligned}$$

Example 1.10 (Einstein's equations)

$$\operatorname{Ric}(g) - \frac{1}{2} g R(g) = 0$$

1.2 Data and Well-Posedness

All of the examples from above need additional information to solve, which we call the *data*. For example, we might need $u|_{\partial\mathcal{U}}$ and so on. A guiding principle to this process is called *well-posedness (in the sense of Hadamard)*.

We say that a PDE problem (equation and the data) is well-posed if we have

1. A solution exists (in some function space).
2. Given some data, the solution should be unique (depends on the function space of choice).
3. The solution depends continuously on the data.

The aim is to find the largest space for which a solution exists, but small enough so that it is unique. (For example, strong cosmic censorship in GR?)

Notation 1.11 (Multi-index notation)**.** We will use multi-index notation,

$$\mathbb{N} = \{0, 1, \dots\}$$

$\alpha = (\alpha_1, \dots, \alpha_n) \in \mathbb{N}^n$ is called a *multi-index*, and we define the *order* of α

$$|\alpha| = \alpha_1 + \alpha_2 + \cdots + \alpha_n$$

and the α-th derivative is

$$D^\alpha f(x) = \partial_1^{\alpha_1} \cdots \partial_n^{\alpha_n} f$$

If $x = (x_1, \dots, x_n)$, then $x^\alpha = x_1^{\alpha_1} \cdots x_n^{\alpha_n}$, and

$$\alpha! = \alpha_1! \cdots \alpha_n!$$

1.3 Classifying PDEs (of order k)

We say eq. (1) is *linear* if F is a linear function of x, u and its derivative. That is, we can write it as

$$\sum_{|\alpha| \le k} a_\alpha(x) D^\alpha u = f(x)$$

Moreover, we say that a linear PDE is *homogeneous* if $f = 0$. We say eq. (1) is *semilinear* if the highest order derivatives appear linearly with coefficients depending only on x. That is, we have

$$\sum_{|\alpha|=k} a_\alpha(x) D^\alpha u(x) + F(x, u, Du, \dots, D^{k-1}u) = 0$$

One example would be $\Delta u = u_x^2$. Finally, we say eq. (1) is *quasilinear* if the highest order derivatives appear linearly, but the coefficients depending on lower order derivatives. That is, we have

$$\sum_{|\alpha|=k} a_\alpha(x, u, \dots, D^{k-1}u) D^k u + F(x, u, \dots, D^{k-1}u) = 0$$

For example, we can have $uu_{xx} + u_{yy} - u_x^2 = 0$. Finally, we say eq. (1) is *fully non-linear* if it is non of the above.

2 Cauchy-Kovalevskaya theorem

2.1 ODE theory

Fix $\mathcal{U} \subseteq \mathbb{R}^n$ open, and suppose $f : \mathcal{U} \to \mathbb{R}^n$ is given. We would like to consider the ODE

$$\begin{cases} \dot{u}(t) = f(u(t)) \\ u(0) = u_0 \end{cases} \tag{2}$$

Theorem 2.1 (Picard-Lindelöf)**.** Suppose we have $r, K > 0$ such that $B_r(u_0) \subseteq U$ and

$$\|f(x) - f(y)\| \leq K\|x - y\|$$

for all $x, y \in B_r(u_0)$. Then there exists $\varepsilon > 0$, depending on K, r and a unique C^1 solution $u : (-\varepsilon, \varepsilon) \to \mathcal{U}$ solving eq. (2).

Lecture 2

Sketch proof, see examples sheet 1. If $u \in \mathbb{C}^1$ solves eq. (2), then by the fundamental theorem of calculus, u satisfies the *weak formulation*

$$u(t) = u_0 + \int_0^t f(u(s))\mathrm{d}s \tag{3}$$

Moreover, if $u \in C^0$ is a solution to eq. (3), then it is a C^1 solution to eq. (2). Moreover, if u exists, then it is a fixed point of

$$G(w) = u_0 + \int_0^t f(w(s))\mathrm{d}s$$

Let $\mathcal{S} = \left\{w : (-\varepsilon, \varepsilon) \to \overline{B_{r/2}(u_0)}\right\}$ continuous. We want to show that $\mathcal{S}$ is a complete metric space, $G : \mathcal{S} \to \mathcal{S}$ is a contraction for ε sufficiently small, and by the contraction mapping theorem G has a fixed point. □

Remark 2.2.

1. The solution (in general) can't be global. Consider for example
$$\dot{u}(t) = u(t)^2 \quad \text{with} \quad u(0) = u_0 > 0$$
Solutions to this equation blow up in finite time.
2. This does not apply to
$$\dot{u}(t) = \sqrt{u(t)} \quad \text{with} \quad u(0) = 0$$
There are two solutions. Note we *can* apply the Peano existence theorem.

Now suppose f is smooth, and we have $\dot{u}(t) = f(u(t))$ is C^1. By the chain rule,

$$\ddot{u}(t) = Df(u(t)) \cdot \dot{u}(t) = f_2(u(t), \dot{u}(t))$$

which is continuous. Hence $\ddot{u}$ is continuous, and so $u \in C^2$. Repeating this, we get that $u \in C^k$ for all k. That is, u is smooth.

In principle, given $u_0 = u(0)$, we can determine

$$u^{(k)}(0) = F_k\left(u, u', \ldots, u^{(k-1)}\right)\Big|_{t=0}$$

and so we can write

$$\sum_{k\geq 0} \frac{u^{(k)}(0)}{k!} t^k$$

We call this a *formal power series solution*. Does our solution $u(t)$ agree with this? That is, do we have

$$u(t) = \sum_{k\geq 0} \frac{u^{(k)}(0)}{k!} t^k$$

in a neighbourhood of 0?

Theorem 2.3 (Cauchy-Kovalevskaya for simple ODEs). If $f(u)$ is real analytic in a neighbourhood of u_0, then the series

$$\sum_{k\geq 0} \frac{u^{(k)}(0)}{k!} t^k$$

converges in a neighbourhood of 0 to the unique solution of eq. (2) given by Picard-Lindelöf.

2.2 Real analyticity and majorants

Suppose $f : (-\varepsilon, \varepsilon) \to \mathbb{R}$ is a smooth function. Therefore, $f^{(n)}(0)$ exists for all $n \geq 0$. Does the partial sums

$$\sum_{n\geq 0} \left|f^{(n)}(0)\right| n! x^n$$

for some $|x| \leq \delta$? No, consider the function

$$f(x) = \begin{cases} \exp\left(-\frac{1}{x}\right) & x > 0 \\ 0 & x \leq 0 \end{cases}$$

This is a smooth function, with $f^{(n)}(0) = 0$ for all n.

Definition 2.4 (real analytic)
Let $\mathcal{U} \subseteq \mathbb{R}^n$ be open and $f : \mathcal{U} \to \mathbb{R}$ is *real analytic* (at x_0) if there exists $r > 0$, $f_\alpha \in \mathbb{R}$ such that

$$f(x) = \sum_{\alpha} f_\alpha (x - x_0)^\alpha$$

when $|x - x_0| < r$.

Remark 2.5.

1. That is, f can be written as a convergent power series and
$$f_\alpha = \frac{D^\alpha f(x_0)}{n!}$$
2. Real analyticity is a local property.
3. f is real analytic on an open set U if it is real analytic at each $x_0 \in U$.
4. We will denote the set of real analytic functions on $\mathcal{U}$ by $C^\omega(\mathcal{U})$.
5. If f is C^ω, then f is smooth (e.g. Weierstrass M-test).
6. If f is real analytic, and $\mathcal{U}$ is connected, then f is uniquely determined in $\mathcal{U}$ by its derivatives $D^\alpha f(x)$ at some point $x \in \mathcal{U}$.

7. In particular, f is real analytic if and only if for any compact $K \subseteq U$, there exists C, r such that

$$\sup_{x \in K} |D^\alpha f(x)| \leq C|\alpha|! r^{|\alpha|}$$

Exercise: Show $f(x) = 1/x$ and $f(x) = \sqrt{x}$ are real analytic for $x > 0$.

Example 2.6

Recall

$$\frac{1}{1-x} = \sum_{k \geq 0} x^k$$

for $|x| < 1$. Let $r > 0$, and consider

$$f(x) = \frac{r}{r - (x_1 + \cdots + x_n)} = \frac{1}{1 - \frac{x_1 + \cdots + x_n}{r}} = \sum_{k \geq 0} \left(\frac{x_1 + \cdots + x_n}{r} \right)^k$$

provided $|x_1 + \cdots + x_n| \leq \sqrt{n} \left(\sum_j |x_j|^2 \right)^{1/2} = \sqrt{n}\|x\| < r$. By the multinomial theorem (sheet 1),

$$f(x) = \sum_{k \geq 0} \frac{1}{r^k} \sum_{|\alpha| = k} \binom{|\alpha|}{\alpha} x^\alpha = \sum_\alpha \frac{|\alpha|!}{\alpha! r^{|\alpha|}} x^\alpha$$

where

$$\binom{|\alpha|}{\alpha} = \frac{|\alpha|!}{\alpha!}$$

and so,

$$f(x) = \sum_\alpha f_\alpha x^\alpha$$

where

$$f_\alpha = \frac{|\alpha|!}{\alpha!} \frac{1}{r^{|\alpha|}}$$

This series is absolutely convergent near zero, since

$$\sum_\alpha \frac{|\alpha!|}{\alpha!} \frac{x^\alpha}{r^{|\alpha|}} = \sum_{k \geq 0} \left(\frac{|x_1| + \cdots + |x_n|}{r} \right)^k < \infty$$

Definition 2.7 (majorise)

Let $f = \sum_\alpha f_\alpha x^\alpha$ and $g = \sum_\alpha g_\alpha x^\alpha$, we say that g *majorises* f, or g *is a mojorant of* f, written $g \gg f$, if $g_\alpha \geq |f_\alpha|$ for all α.

For vector valued functions, we require each component to be a majorant.

Lemma 2.8 (properties of majorants).

1. If $g \gg f$, and g converges for $\|x\| < r$, then f converges for $\|x\| < r$.
2. If $f = \sum f_\alpha x^\alpha$ converges for $\|x\| < r$, then for any $s \in (0, r/\sqrt{n})$, there exists a majorant of f which converges for $\|x\| < s/\sqrt{n}$.

Proof. 1. Looking at the partial sums

$$\sum_{|\alpha| \leq k} |f_\alpha x^\alpha| = \sum_{|\alpha| \leq k} |f_\alpha| |x_1|^{\alpha_1} \cdots |x_n|^{\alpha_n} \leq \sum_{|\alpha| \leq k} |g_\alpha| |x_1|^{\alpha_1} \cdots |x_n|^{\alpha_n} \leq \sum_\alpha |x_1|^{\alpha_1} \cdots |x_n|^{\alpha_n} = g(\tilde{x})$$

where $\tilde{(x)} = (|x_1|, \ldots, |x_n|)$. So $\left\|\tilde{x}\right\| = \|x\|$, and so if $\|x\| < r$, then g converges at $\tilde{x}$. That is, $g(\tilde{x}) < \infty$. Therefore, we have a uniform bound on the partial sum.

2. Let $s \in (0, r/\sqrt{n})$, and set $y = (s, \dots, s)$. Then $\|y\| = s\sqrt{n}$, and by assumption,

$$f(y) = \sum_{\alpha} f_\alpha y^\alpha$$

converges, as $\|y\| = s\sqrt{n} < r$. So there exists a constant c such that $|f_\alpha y^\alpha| \leq c$. Hence

$$|f_\alpha| \leq \frac{C}{|y^\alpha|} = \frac{C}{|y_1|^{\alpha_1} \cdots |y_n|^{\alpha_n}} = \frac{C}{|s|^{|\alpha|}} \leq \frac{C}{|s|^{|\alpha|}} \frac{|\alpha|!}{\alpha!}$$

So we define

$$g(x) = \frac{Cs}{s - (x_1 + \cdots + x_n)}$$

From the above, g majorises f. □

Lecture 3

2.3 Proof of Cauchy-Kovalevskaya for ODEs

Proof of theorem 2.3. We will use the method of majorants. Without loss of generality $u_0 = 0$, and for simplicity, we can assume $n = 1$. We need to find the series coefficients. So

$$\dot{u} = f(u)$$

and so $\dot{u}(0) = f(u(0)) = f(0)$, that is, $u_1 = f(0)$. Next,

$$\ddot{u}(t) = f'(u(t))\dot{u}(t)$$

and so $\ddot{u}(0) = f'(0)f(0)$, that is, $u_2 = f'(0)f(0) = \frac{1}{2!}f'(0)f(0)$. Repeating,

$$u^{(3)}(0) = f''(0)f(0)^2 + f'(0)^2 f(0)$$

and so

$$u_3 = \frac{1}{3!}(f''(0)f(0)^2 + f'(0)^2 f(0))$$

Iterating this procedure,

$$u_k = P_k\left(f(0), \dots, f^{(k-1)}(0)\right)$$

where P_k is a polynomial in k-variables, with nonnegative coefficients. For example,

$$P_1(x) = x$$
$$P_2(x, y) = \frac{1}{2!}xy$$
$$P_3(x, y, z) = \frac{1}{3!}(x^2 z + xy^2)$$

Since f is real analytic, we have that

$$f(v) = \sum_{k \geq 0} f_k v^k$$

where $f_k = \frac{1}{k!}f^{(k)}(0)$. Hence we have that

$$f^{(k)}(0) = k! \cdot f_k$$

Substituting, we have that

$$u_k = Q_k(f_0, \dots, f_{k-1})$$

which again is a polynomial in k-variables and nonnegative coefficients. This polynomial is "universal".

Aim: We would like to show that the power series

$$\sum_k u_k t^k$$

converges in a neighbourhood of $t = 0$, and solves the ODE, eq. (2). Since f is analytic, we know that

$$f(u) = \sum_k f_k u^k$$

on $|u| < k$. Fixing some $s < r$, there exists a majorant

$$g(u) = \sum_k g_k u^k$$

of f, from lemma 2.8 (ii). Consider the auxilliary differential equation

$$\dot{w}(t) = g(w(t))$$

and $w(0) = 0$. Using the definition of g, we that

$$\frac{\mathrm{d}w}{\mathrm{d}t} = \frac{Cs}{s - w(t)}$$

We get that

$$w = s \pm \sqrt{s^2 - 2Cst}$$

Due to the initial data, we take the $-$ solution. That is,

$$w = s - \sqrt{s^2 - 2Cst}$$

This is real analytic, for $|t| < s/2C$. This tells us that

$$w(t) = \sum_k w_k t^k$$

converges for $|t| < s/2C$. Moreover,

$$w_k = Q_k(g_0, \dots, g_{k-1})$$

since Q_k is "universal".

Claim 2.9. w majorises u.

By construction, g majorises f, i.e. $g_k \geq |f_k|$ for all k. Moreover, since Q_k has nonnegative coefficients,

$$w_k = Q_k(g_0, \dots, g_{k-1}) \geq Q_k(|f_0|, \dots, |f_{k-1}|) \geq |Q_k(f_0, \dots, f_{k-1})| = |u_k|$$

Hence by lemma 2.8 (i), we know that the series

$$\sum_k u_k t^k$$

converges for $|t| < s/2C$.

To conclude, set

$$u(t) := \sum_{k \geq 0} u_k t^k$$

and we need to check that it solves eq. (2). Both sides are analytic, so suffices to check the derivatives on each side agree to all orders at $t = 0$. □

Remark 2.10.

1. We can extend to systems, where we replace u_k with

$$u_k^j = Q_k^j(D^{|\alpha|} \mid |\alpha| \leq k)$$

   For w, we can replace $w^j = w^1$ as before.

2. For the non-autonomous case,

$$u(t) = f(u, t)$$
$$u(0) = 0$$

Consider $v(t) = (u(t), t)$, then $\dot{v}(t) = (\dot{u}(t), 1) = (f(u, 1), 1) = (f(v), 1) = F(v)$ with $v(0) = 0$, and we can apply the system version.

For the PDE version, see the handout.

2.4 Cauchy-Kovalevskaya for PDEs

Let $u : \mathbb{R}^n \to R^m$, and choose $r > 0$. Consider the following problem

$$u_t = \sum_{j=1}^{n} B_j(u, x) u_{x_j} + C(u, x)$$

on $||x||^2 + t^2 < r^2$, with

$$u = 0$$

on $||x||^2 + t^2 < r^2$ and $t = 0$. The B_j are matrices, B_j and C are real analytic.

Theorem 2.11 (Cauchy-Kovalevskaya for first order systems)**.** Suppose B_j, C are real analytic, for small $r > 0$. Then there exists a unique real analytic function

$$u = \sum_{\alpha} u_\alpha x^\alpha$$

solving the above PDE.

Idea. Compute

$$u_\alpha = \frac{D^\alpha u}{\alpha!}$$

in terms of B_j, C, and show that the power series converges for small r. We use the PDE to find all derivatives. □

Example 2.12

Consider the system

$$u_t = v_x - f$$
$$v_t = -u_x$$

with $u = v = 0$ on $t = 0$. The boundary conditions give us that

$$u(0, 0) = v(0, 0) = 0$$

We would like to determine u_α for all α. By differentiating the boudary conditions,

$$\partial_x^n u(x, 0) = \partial_x^n v(x, 0) = 0$$

for all n. That is, for the case $\alpha = (n, 0)$. From the PDE,

$$u_t(x, 0) = 0 - f = -f \qquad v_t(x, 0) = 0$$

This then means that

$$\partial_x^n \partial_t u(x, 0) = -\partial_x^n f(x, 0)$$

and

$$\partial_x^n \partial_t v(x, 0) = 0$$

for all $n \geq 1$.

Next, if $\alpha = (n, 2)$ use the PDE and we get

$$u_{tt}(x, 0) = f_t(x, 0)$$

and

$$v_{tt}(x,0) = f_x(x,0)$$

The same method as above gives us that

$$\partial_x^n \partial_t^2 u(x,0) = -(\partial x)^n \partial_t f(x,0)$$
$$\partial_x^n \partial_t^2 v(x,0) = (\partial_x)^{n+1} f(x,0)$$

Repeating this, we can compute all of the derivatives.

Lecture 4

2.5 Reduction to first order systems

Example 2.13

Consider $u : \mathbb{R}^3 \to \mathbb{R}$ satisfying

$$u_{tt} = uu_{xy} - u_{xx} + u_t$$

with conditions

$$u|_{t=0} = u_0(x,y) \quad \text{and} \quad u_t|_{t=0} u_1(x,y)$$

where u_0, u_1 are real analytic near $0 \in \mathbb{R}^3$. Note

$$f(t,x,y) = u_0 + tu_1$$

is real analytic near $0 \in \mathbb{R}^3$. Note

$$f|_{t=0} = u_0 \quad \text{and} \quad \partial_t f|_{t=0} = u_1$$

Set $w(t,x,y) = u - f$. Then we find that

$$w_t t = ww_{xy} - w_{xx} + w_t + fw_{xy} + f_{xy}w + F$$

where

$$R = ff_{xy} - f_{xx} + f_t$$

where

$$w|_{t=0} = \partial_t w|_{t=0} = 0$$

Observe F is real analytic, and independent of w and its derivatives. Let $x = (x,y,t) = (x^1, x^2, x^3)$, and set

$$v = (w, w_x, w_y, w_z) = (v^1, v^2, v^3, v^4)$$

Then

$$\begin{aligned} v_t^1 &= w_t = v^4 \\ v_t^2 &= w_{xt} = v_{x_1}^4 \\ v_t^3 &= w_{yt} = v_{x_2}^4 \\ v_t^4 &= v^1 v_{x_2}^2 - v_{x_1}^2 + v^4 + f v_{x_2}^2 + f_{xy} v^1 + F \end{aligned}$$

Define

$$B_1 = \begin{pmatrix} & & & 0 \\ & & & 1 \\ & & & 0 \\ 0 & -1 & 0 & 0 \end{pmatrix} \qquad B_2 = \begin{pmatrix} & & & \\ & & & \\ & & & 1 \\ v^1 + f & 0 & 0 & 0 \end{pmatrix}$$

and

$$c = \begin{pmatrix} v^4 \\ 0 \\ 0 \\ v^4 + f_{xy} v^1 + F \end{pmatrix}$$

We can see then

$$\frac{\partial}{\partial x_3} v = B_1 v_{x_1} + B_2 v_{x_2} + c$$

In this case, B_1, B_2, c are real analytic functions of x, v, and so we can apply Cauchy-Kovalevskaya.

More generally, consider the scalar quasilinear problem

$$\sum_{|\alpha|=k} a_\alpha (D^{k-1}u, \dots, u, x) D^\alpha u + a_0 (D^{k-1}u, \dots, u, x) = 0$$

where

$$u : B_r(0) \to \mathbb{R} \quad \text{and} \quad \frac{\partial u}{\partial x_n} = \cdots = \left(\frac{\partial}{\partial x_n}\right)^{k-1} u = 0$$

for $\|x'\| < r, x_n = 0$.

Define

$$v = \left(u, \frac{\partial u}{\partial x_1}, \dots, \frac{\partial u}{\partial x_n}, \frac{\partial^2 u}{\partial x_1 \partial x_1}, \dots, \left(\frac{\partial}{\partial x_n}\right)^{k-1} u\right) = (v^1, \dots, v^m) \in \mathbb{R}^m$$

We would like to get a first order system in v. That is, express

$$\frac{\partial v^j}{\partial x_n}$$

in terms of v^j and $\frac{\partial v}{\partial x_j}$ for $j = 1, \dots, m-1$. If $j = 1$, then

$$\frac{\partial v^1}{\partial x_n} = \frac{\partial u}{\partial x_n} = v^\ell$$

for some ℓ. If $2 \le j \le m-1$, then

$$v^j = D^\alpha u$$

for some $|\alpha| \le k-1$, such that $\alpha_n < k-1$. So

$$\frac{\partial v^j}{\partial x_n} = D^\alpha \frac{\partial u}{\partial x_n} = \frac{\partial^{|\alpha|+1}}{\partial_1^{\alpha_1} \cdots \partial_n^{\alpha_n+1}} u$$

If $|\alpha| \le k-2$, then $|\alpha| + 1 \le k-1$,a nd so $\frac{\partial v^j}{\partial x_n} = v^\ell$ for some ℓ.

If $|\alpha| = k-1$, and $\alpha_n < k-1$, then there exists $p \ne n$ such that $\alpha_p \ge 1$. So we have that

$$\frac{\partial v^j}{\partial x_n} = \frac{\partial}{\partial x_n}\left(\frac{\partial^{|\alpha|} u}{\partial_1^{\alpha_1} \cdots \partial_n^{\alpha_n}}\right) = \frac{\partial}{\partial x_p}\left(\frac{\partial^{|\alpha|} u}{\partial_1^{\alpha_1} \cdots \partial_p^{\alpha_p - 1} \cdots \partial_n^{\alpha_n - 1}}\right) = \frac{\partial v^\ell}{\partial x_p}$$

for some ℓ. Finally, to compute

$$\frac{\partial v^m}{\partial x_n}$$

we will use the PDE. Recall the coefficients are $a_\alpha(v, x)$ for $v \in \mathbb{R}^m, x \in \mathbb{R}^n$. We assume $a_\alpha : B_\rho(0) \to \mathbb{R}$ is real analytic, and suppose

$$a_c = a_{(0,\dots,0,k)}(0) \ne 0$$

Since a_α are real analytic near zero, they are continuous. Therefore, $a_c(z, w) \ne 0$ for all $\|z\|^2 + \|w\|^2 \le \delta^2$, where $\delta < \rho$. Then

$$a_c \frac{\partial^k u}{\partial x_n^k} = -\left(\sum_{|\alpha|=k, \alpha_n < k} a_\alpha D^\alpha u + a_0\right)$$

Dividing by a_c, we get

$$\frac{\partial^k u}{\partial x_n^k} = -\frac{1}{a_c}\left(\sum_{|\alpha|=k, \alpha_n < k} a_\alpha D^\alpha u + a_0\right)$$

The right hand side can be written in terms of

$$\frac{\partial v^\ell}{\partial x_p}, v$$

for $p < n$. Therefore, if $a_c \ne 0$, we have turned the scalar quasilinear PDE into a first order PDE system, on which we can apply Cauchy-Kovalevskaya.

Definition 2.14

If $a_c = a_{(0,\dots,0,k)}(0)$ is non-zero, then we say the plane

$$\{x^n = 0\}$$

is *non-characteristic*. Otherwise, it is *characteristic*.

2.6 Exotic boundary conditions

Definition 2.15 (real analytic hypersurface)

We say $\Sigma \subseteq \mathbb{R}^n$ is a *real analytic hypersurface near* $x_0 \in \Sigma$ if there exists $\varepsilon > 0$, and a real analytic function

$$\Phi : B_\varepsilon(x_0) \to U \subseteq \mathbb{R}^n$$

where U is an open neighbourhood of $0 \in \mathbb{R}^n$, and defining $y = \Phi(x)$, with $\Phi(x_0) = 0$. Moreover, we require

(i) Φ is a bijection,

(ii) $\Phi^{-1} : U \to B_\varepsilon(x_0)$ is real analytic,

(iii) $\Phi(\Sigma \cap B_\varepsilon(x_0)) = \{y_n = 0\} \cap U$.

We can think of Φ as "straightening out" Σ.

Example 2.16

Spheres, planes, tori, etc. are real analytic (hyper)surfaces.

Let γ be unit normal to Σ, and consider

$$\sum_{|\alpha|=k} a_\alpha(D^{k-1}u, \dots, u, x)D^\alpha u + a_0(D^{k-1}u, \dots, u, x)$$

where

$$u = \sum_i \gamma^i \partial_i u = (\gamma^i \partial_i)^{k-1} u = 0 \text{ on } \Sigma \tag{4}$$

Define

$$v(y) = u(\Phi^{-1}(y))$$

for $y \in U$. That is, $u(x) = v(\Phi(x))$ for $x \in B_\varepsilon(x_0)$. Using the chain rule,

$$\frac{\partial u}{\partial x_i} = \sum \frac{\partial u}{\partial y_j}\frac{\partial \Phi^j}{\partial x_i}$$

where $\Phi = (\Phi^1, \dots, \Phi^n) \in \mathbb{R}^n$. So the PDE becomes

$$\sum_{|\alpha|=k} b_\alpha D^\alpha v + b_0 = 0$$

where b_0, b_α dependis on u and $D^\alpha u$ for $|\alpha| \le k-1$, and also Φ (which is given). The boundary conditions becomes

$$v = \frac{\partial}{\partial y_n} v = \cdots = \left(\frac{\partial}{\partial y_n}\right)^{k-1} v = 0$$

on $\{y_n = 0\}$. Since Φ is real analytic, so are b_0, b_α.

Lecture 5

We would like to apply Cauchy-Kovalevskaya, therefore we need to check that whether the hypersurface $\{y_n = 0\}$ is non-characteristic. That is,

$$b_{(0,\dots,0,k)}(D^{k-1}v = 0, \dots, Dv = 0, y = 0) \neq 0$$

Note if $|\alpha| = 2$, we can compute

$$D^\alpha u = \frac{\partial^k v}{\partial y_n^k}(D\Phi^n)^\alpha + \text{terms not involving} \frac{\partial^k v}{\partial y_n^k}$$

For example, if $k = 2, n = 2, \alpha = (0, 2)$, then

$$D^\alpha u = u_{x_2x_2} = v_{y_2y_2} \underbrace{(\Phi^2_{x_2})(\Phi^2_{x_2})}_{=(D\Phi^2)^\alpha} + \text{terms not involving} v_{y_2y_2}$$

Thus,

$$b_{(0,\dots,0,k)} = \sum_{|\alpha|=k} a_\alpha \, (D\Phi^n)^\alpha$$

Definition 2.17

We say that Σ is *non-characteristic at* $x_0 \in \Sigma$ if

$$b_{(0,\dots,0,k)}(0) = \sum_{|\alpha|=k} a_\alpha(0, \dots, 0, x_0) \, (D\Phi^n(x_0))^\alpha \neq 0$$

Otherwise, Σ is characteristic at x_0.

Remark 2.18. Note that $\Sigma = \left\{x \in \mathbb{R}^n \mid \Phi^{(n)}(x) = y_n = 0\right\}$. This tells us that

$$D\Phi^n(x) = c(x)\gamma(x)$$

where γ is the unit normal of Σ. In particular,

$$D\Phi^n(x_0) = c(x_0)\gamma(x_0)$$

and so the non-characteristic condition is equivalent to

$$\sum_{|\alpha|=k} a_\alpha \gamma^\alpha(x) \neq 0$$

Theorem 2.19 (Cauchy-Kovalevskaya for non-characteristic hypersurfaces). Suppose $\Sigma \subseteq \mathbb{R}^n$ is a hypersurface, with normal γ, and consider the PDE eq. (4) as above. Suppose a_α, a_0 are real analytic near $x_0 \in \Sigma$, and Σ is non-characteristic near x_0. Then there exists a unique real analytic solution in a neighbourhood of x_0.

2.7 Characteristic surfaces

Consider the linear operator

$$L = \sum_{i,j=1}^n a_{ij} \frac{\partial^2}{\partial x_i \partial x_j}$$

where $a_{ij} \in \mathbb{R}$. Without loss of generality, we can assume $a_{ij} = a_{ji}$. Consider the PDE problem

$$\begin{cases} Lu = f \\ u = \gamma^i \partial_i u = 0 \quad \text{on } \Pi_\gamma = \{x \mid x \cdot \gamma = 0\} \end{cases} \tag{5}$$

That is, the boundary conditions are on the plane with unit normal γ. In particular, Π_γ is non-characteristic for eq. (5) if

$$\sum_{i,j=1}^n a_{ij}\gamma^i\gamma^j \neq 0$$

We would like to find non-characteristic Π_γ. Note that the left hand side is just $\langle A\gamma, \gamma\rangle$, where $A = (a_{ij})$ is a symmetric matrix, with the usual Euclidean inner product. In particular, A is diagonalisable, say $A = P^{\mathsf{T}}\Lambda P$, where P is orthogonal and Λ is diagonal. Then

$$\langle A\gamma, \gamma\rangle = \left\langle P^{\mathsf{T}}\Lambda P\gamma, \gamma\right\rangle = \langle \Lambda v, v\rangle$$

where $v = P\gamma$. If $\{\lambda_i\}$ are the eigenvalues for A, then the non-characteristic condition becomes

$$\sum_{i=1}^{n} \lambda_i (v_i)^2 \neq 0$$

Example 2.20 (Laplacian)

$$L = \Delta = \sum_{i=1}^{n} \partial_{1,1}^2$$

gives

$$A = \begin{pmatrix} 1 & & \\ & \ddots & \\ & & 1 \end{pmatrix}$$

The Laplacian is an elliptic operator.

Example 2.21 (Wave equation)

$$L = \partial_t^2 + \Delta$$

gives

$$A = \begin{pmatrix} 1 & & & \\ & \ddots & & \\ & & 1 & \\ & & & -1 \end{pmatrix}$$

Case 1: All eigenvalues have the same sign (and are all non-zero). Since v is a unit vector, the characteristic condition is impossible. That is, there are no characteristic hyperplanes Π_γ. In this case, we call L an *elliptic operator*.

Case 2: Say $\lambda_n < 0$ and $\lambda_j > 0$ for $j \neq n$ (or vice versa). In this case, we call L an *hyperbolic operator*. In particular, the characteristic condition becomes

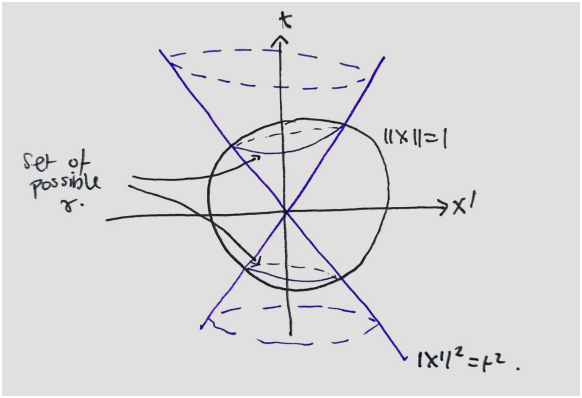
$$\sum \lambda_i v_i^2 = 0 \iff \lambda_1 v_1^2 + \sum_{j=2}^{n} \lambda_j v_j^2 = 0$$

Considering the wave equation again, we have the condition that

$$v_n^2 = \sum_{j=1}^{n-1} v_j^2$$

subject to the condition

$$||v|| = 1$$



Note that these cases are *not* exhaustive.

Now we want to different features of elliptic and hyperbolic operators. We will forget about boundary conditions, and look for solutions of the form

$$u(x) = e^{ik\cdot x}$$

for $k \in \mathbb{R}^n$. We are looking for *wave-like* solutions. Substituting,

$$L(e^{ik\cdot x}) = e^{ik\cdot x} = \sum_{j,\ell} a_{j\ell} k_j k_\ell$$

We would like to consider $Lu = 0$. Taking $k = c\gamma$, $\|\gamma\| = 1$, then the condition is equivalent to

$$c^2 \sum a_{j\ell}\gamma_j\gamma_\ell = 0$$

If L is elliptic, then the only solution is when $k = 0$. That is, there are no wave-like solutions. On the other hand, if L is hyperbolic, then we can have wave like solutions, that is,

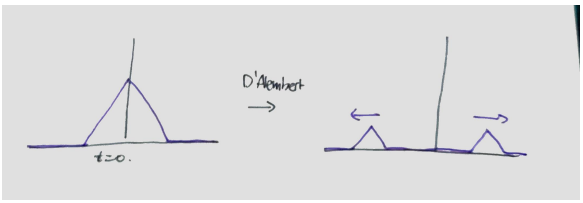
$$\sum a_{ij}\gamma_i\gamma_j = 0$$

has a solution with $\|\gamma\| = 1$. In this case, we have

$$u(x) = e^{i\lambda\gamma\cdot x}$$

gives an infinite family of solutions, indexed by $\lambda \in \mathbb{R}$.

As we take $|\lambda| \to \infty$, we see that $u'(x)$ can grow large. In particular, solutions can be rough.



By contrast, we will see that solutions to elliptic equations are smooth.

Lecture 6

Example 2.22

Consider the IVP for Laplace's equation

$$\begin{cases} u_{xx} + u_{yy} = 0 \\ u(x,0) = \varphi(x) \\ \partial_y u(x,0) = 0 \end{cases}$$

Is this problem well-posed? If $\varphi(x) = 0$, then 0 is a solution. On the other hand, we don't have Cauchy stability. Consider

$$u_k(x,y) = e^{-\sqrt{k}}\cos(kx)\cosh(ky)$$

See typed notes for more details.

3 Sobolev spaces

3.1 Hölder spaces $C^{k,\gamma}$

Let $U \subseteq \mathbb{R}^n$ be open, $k \in \mathbb{N}$.

Definition 3.1 (C^k spaces)

Define

$$C^k(U) = \{f : U \to \mathbb{R} \mid u \text{ is } k \text{ times continuously differentiable}\}$$

and define

$$C^k(\overline{U}) = \left\{u \in C^k(U) \mid u \text{ and its derivatives are bounded and uniformly continuous on } U\right\}$$

We will define the norm

$$\|u\|_{C^k(\overline{U})} = \sum_{|\alpha| \leq k} \sup_{x \in U} |D^\alpha u(x)|$$

The idea is that $C^k(\overline{U})$ is the space of functions which can be extended continuously to ∂U. Note that this is is contained in, but not equal to

$$u : \overline{U} \to \mathbb{R} \text{ such that } u \text{ and its derivatives are continuous}$$

On examples sheet 2, we will show that $C^k(\overline{U})$ is a Banach space.

Definition 3.2 (Hölder continuous)

We say $u : U \to \mathbb{R}$ is *Hölder continuous of index* γ with $0 < \gamma \leq 1$ if there exists a constant $C > 0$, such that

$$|u(x) - u(y)| \leq C|x - y|^\gamma$$

for all $x, y \in U$.

If $\gamma = 1$, then we say that u is *Lipschitz continuous*.

Remark 3.3. If $\gamma > 1$, and u is Hölder continuous of index γ, then u is constant.

Definition 3.4 (0-Hölder space)

For $\gamma \in (0, 1]$, we define the 0-*Hölder space*:

$$C^{0,\gamma}(\overline{U}) = \left\{u \in C^0(\overline{U}) \;\middle|\; u \text{ is } \gamma\text{-Hölder continuous}\right\}$$

Define the γ-Hölder seminorm by

$$[u]_{C^{0,\gamma}(\overline{U})} = \sup_{x,y \in U} \frac{|u(x) - u(y)|}{|x - y|}$$

That is, the smallest C such that u is γ-Hölder continuous. Since constant functions vanish, we add the C^0 norm, and define

$$\|u\|_{C^{0,\gamma}(\overline{U})} = \|u\|_{C^0(\overline{U})} + [u]_{C^{0,\gamma}(\overline{U})}$$

The space $C^{0,\gamma}$, with the Hölder norm $\|\cdot\|_{C^{0,\gamma}(\overline{U})}$ is a Banach space. We can extend this to higher order, that is,

Definition 3.5 (k-th Hölder space)
Define the *k-th Hölder space*

$$C^{k,\gamma}(\overline{U}) = \left\{u \in C^k(\overline{U}) \mid D^\alpha u \in C^{0,\gamma}(\overline{U}) \text{ for all } |\alpha| = k\right\}$$

with norm

$$\|u\|_{C^{k,\gamma}(\overline{U})} = \|u\|_{C^k(\overline{U})} + \sum_{|\alpha|=k} [D^\alpha u]_{C^{0,\gamma}(\overline{U})}$$

As above, $C^{k,\gamma}$ is a Banach space with the Hölder norm.

3.2 The Lebesgue spaces

Definition 3.6 (L^p space)
Let $U \subseteq \mathbb{R}^n$ be open, and suppose $1 \leq p \leq \infty$, define

$$L^p(U) = \frac{\left\{f : U \to \mathbb{R} \mid f \text{ measurable, and with} \|u\|_{L^p(U)} < \infty\right\}}{\sim}$$

where

$$\|u\|_{L^p} = \begin{cases} \left(\int_U |u(x)|^p \mathrm{d}x\right)^{1/p} & 1 \leq p < \infty \\ \operatorname{ess\,sup}_{x \in U} |u(x)| = \inf\{C > 0 \mid |u(x)| \leq C \text{ a.e.}\} & p = \infty \end{cases}$$

and $u \sim v$ if $u = v$ almost everywhere.

$L^p(U)$ with the $\|\cdot\|_{L^p}$ norm is a Banach space. We also define local versions,

Definition 3.7 (L^p_{loc} space)
We say that $u \in L^p_{loc}(U)$ if $f \in L^p(V)$ for every $V \Subset U$ ($V \subseteq K \subseteq U$, where K is compact). Equivalently,

$$L^p_{loc}(U) = \bigcap_{V \Subset U} L^p(V)$$

Note that $L^p_{loc}(U)$ is not Banach, on the other hand, it is a Fréchet space.

Remark 3.8. If $K \subseteq U$ is compact, U is open, then

$$d(K, \partial U) = \inf\{|x - y| \mid x \in K, y \in \mathbb{R}^n \setminus U\} > 0$$

We will use the space outside K as a "buffer zone".

3.3 Weak derivatives

That is, a notion of derivative for L^p.

Definition 3.9 (weak derivative)
Suppose $u, v \in L^1_{loc}(U)$, α a multi-index. We say v is the *α-th weak derivative of U* if

$$\int_U u D^\alpha \phi \mathrm{d}x = (-1)^{|\alpha|} \int_U v \phi \mathrm{d}x$$

for all $\phi \in C_c^\infty(U)$. We will also call the space $C_c^\infty(U)$ the space of *test functions*.

Remark 3.10. 1. since $\operatorname{supp}(D^\alpha \phi)$ and $\operatorname{supp}(\phi)$ are compact, the integrals are finite,

2. u, v obey the correct integration by parts formula

Example 3.11

$u(x) = |x|$ is not differentiable at $x = 0$, but it is weakly differentiable with $v(x) = \operatorname{sign}(x)$.

Lemma 3.12 (uniqueness of weak derivative)**.** Suppose $v, \tilde{v} \in L^1_{loc}(U)$ are both the α-th weak derivative of $u \in L^1_{loc}(U)$, then $v = \tilde{v}$ almost everywhere.

Proof. For all $\varphi \in C_c^\infty(U)$,

$$\int_U v\phi \mathrm{d}x = (-1)^{|\alpha|} \int_U u D^\alpha \phi \mathrm{d}x = \int_U \overline{v}\phi \mathrm{d}x$$

Thus,

$$\int_U (v - \overline{v})\phi \mathrm{d}x = 0$$

for all $\phi \in C_c^\infty(U)$. Taking ϕ to be a smooth approximation to an indicator, we get the required result. □

Suppose u is smooth, then the weak derivative agrees with the usual derivative almost everywhere.

Notation 3.13. We will write $v = D^\alpha u$.

Definition 3.14 (Sobolev space)

Define the *Sobolev space*

$$W^{k,p}(U) = \left\{ u \in L^1_{loc}(U) \mid u \in L^p(U), \text{ the weak derivatives } D^\alpha u \text{ exists for } |\alpha| \leq k,\ D^\alpha u \in L^p(U) \right\}$$

with the *Sobolev norm*

$$\|u\|_{W^{k,p}(U)} = \begin{cases} \left(\sum_{|\alpha|=k} \int_U |D^\alpha u|^p \mathrm{d}x \right)^{1/p} & p < \infty \\ \sum_{|\alpha| \leq k} \operatorname{ess\,sup}_U |D^\alpha u| & p = \infty \end{cases}$$

When $p = 2$, we write $H^k = W^{k,2}$.

Definition 3.15 ($W_0^{k,p}$)

We denote by $W_0^{k,p}(U)$ the completion of $C_c^\infty(U)$ with respect to the $W^{k,p}$ norm. Analogously, we define $H_0^k = W_0^{k,2}(U)$.

The $_0$ denotes that the function vanishes on the boundary.

Example 3.16

Let $n > 2, \lambda > 0$, and take $U = B_1(0) \subseteq \mathbb{R}^n$ the open ball. Consider

$$u(x) = \begin{cases} |x|^{-\lambda} & \text{for } x \neq 0 \\ \text{anything} & x = 0 \end{cases}$$

When is $u \in W^{p,1}(U)$?

4 Elliptic PDEs

5 Hyperbolic PDEs

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